

作业解答 Homework Solutions

线性代数

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题 Q1

Suppose f is a function and L and a are real numbers.

解:

See sub-problems.

(a) Describe what is meant by $\lim_{x \rightarrow a} f(x) = L$ as best you can.

(包含抽象函数, 需要 LLM)

(b) Graph a function $y = f(x)$ such that $\lim_{x \rightarrow a} f(x) = L$, and $f(a) = L$.

画图题: 请用图形工具。

See graph

(c) Graph a function $y = f(x)$ such that $\lim_{x \rightarrow a} f(x) = L$, but $f(a) \neq L$.

画图题: 请用图形工具。

See graph

(d) Graph a function $y = f(x)$ such that $\lim_{x \rightarrow a} f(x) = L$, and $f(a)$ does not exist.

画图题: 请用图形工具。

See graph

题 Q2

Is ∞ a number? What does $\lim_{x \rightarrow a} f(x) = \infty$ mean? Does $\lim_{x \rightarrow a} f(x) = \infty$ mean that the limit exists?

解:

∞ is **not** a real number.

$\lim_{x \rightarrow a} f(x) = \infty$ means $f(x)$ grows without bound as $x \rightarrow a$ —for every $M > 0$ there is a $\delta > 0$ with $f(x) > M$ whenever $0 < |x - a| < \delta$.

The limit does **not exist** in the usual sense; writing “ $= \infty$ ” describes the *manner of divergence*, not a limit value.

最终答案:

∞ is not a number. $\lim = \infty$ means unbounded growth (DNE), not an existing limit.

题 Q3

What does the Squeeze Theorem say? Draw a picture illustrating this theorem.

解:

Squeeze Theorem: If $g(x) \leq f(x) \leq h(x)$ near a (except possibly at a), and $\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x) = L$, then $\lim_{x \rightarrow a} f(x) = L$.

Intuition: f is squeezed between two functions converging to L .

Application: $\lim_{x \rightarrow 0} x^2 \sin(1/x) = 0$ since $-x^2 \leq x^2 \sin(1/x) \leq x^2$.

最终答案:

If $g(x) \leq f(x) \leq h(x)$ and $\lim g = \lim h = L$, then $\lim f = L$.

题 Q4

What do we mean by $\lim_{x \rightarrow a^-} f(x) = L$? What is meant by $\lim_{x \rightarrow a^+} f(x) = L$? Graph a function $y = f(x)$ and some point $x = a$ where $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$. Find an algebraic expression for such a function.

解:

$\lim_{x \rightarrow a^-} f(x) = L$: **left-hand limit** —approach a from below (through values less than a).

$\lim_{x \rightarrow a^+} f(x) = L$: **right-hand limit** —approach a from above.

Two-sided limit exists $\iff$ both one-sided limits exist and are equal.

Example: $f(x) = \begin{cases} x & x < 0 \\ x + 1 & x \geq 0 \end{cases}$ at $a = 0$: $\lim_{x \rightarrow 0^-} = 0 \neq 1 = \lim_{x \rightarrow 0^+}$, so the limit

DNE.

最终答案:

Left: approach from below; right: from above; two-sided exists iff both agree.

题 P1

(a) Is the limit of a sum always the sum of the limits? Give an example where $\lim_{x \rightarrow a} [f(x) + g(x)]$ exists even though neither $\lim_{x \rightarrow a} f(x)$ nor $\lim_{x \rightarrow a} g(x)$ exist.

解:

No —the **Sum Law** requires both limits to exist.

Counterexample: $f(x) = \frac{1}{x}$, $g(x) = -\frac{1}{x}$ at $x = 0$: neither limit exists, yet $\lim[f+g] = \lim 0 = 0$.

最终答案:

No. $f=1/x$, $g=-1/x$: sum limit = 0 but individual limits DNE.

题 P2

(a) Find $\lim_{x \rightarrow 0^+} (\frac{1}{x} - \frac{1}{|x|})$, $\lim_{x \rightarrow 0^-} (\frac{1}{x} - \frac{1}{|x|})$, and $\lim_{x \rightarrow 0} (\frac{1}{x} - \frac{1}{|x|})$, if they exist.

解:

$$\lim_{x \rightarrow 0^+} -\frac{1}{|x|} + \frac{1}{x} = 0$$

$$\lim_{x \rightarrow 0^-} -\frac{1}{|x|} + \frac{1}{x} = -\infty$$

$$\text{Left: } \lim_{x \rightarrow 0^-} -\frac{1}{|x|} + \frac{1}{x} = -\infty$$

$$\text{Right: } \lim_{x \rightarrow 0^+} -\frac{1}{|x|} + \frac{1}{x} = 0$$

Since $-\infty \neq 0$, the two-sided limit **does not exist**.

$$\text{Left: } \lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1$$

$$\text{Right: } \lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1$$

Since $-1 \neq 1$, the two-sided limit **does not exist**.

(b) Show that $\lim_{x \rightarrow 0} |x| = 0$.

(无法解析子题极限)

(c) Show that $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist.

$$\text{Left: } \lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1$$

$$\text{Right: } \lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1$$

Since $-1 \neq 1$, the limit DNE.

DNE

最终答案:

0, $-\infty$, DNE, DNE

题 P3

For what kinds of functions can one evaluate the limit $\lim_{x \rightarrow a} f(x)$ by just plugging in $x = a$? Give an example of a function $f(x)$ for which the limit $\lim_{x \rightarrow a} f(x)$ exists and can be evaluated, yet for which this substitution doesn't work.

解:

Continuous functions allow direct substitution: $\lim_{x \rightarrow a} f(x) = f(a)$.

Includes polynomials, rational functions (where denominator $\neq 0$), $\sin x$, $\cos x$, e^x , $\ln x$ on their domains.

Counterexample needing algebra first: $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1}$ gives $0/0$ by substitution but $= 2$ after cancellation.

最终答案:

Continuous functions allow direct substitution; $0/0$ forms need algebra first.

题 AP1

Below are listed three limits. Match the best method to each limit and evaluate each limit.

解:

$$\lim_{x \rightarrow 0} \frac{x^{1000} - 703x^{506} + \pi x^{17} + 480}{-372x^{75} + 39x^{14} + \sqrt{7}x^5 - 24} = -20$$

$$\lim_{x \rightarrow 4} \frac{x^2 + 2x - 24}{x - 4} = 10$$

$$\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$$

(a) $\lim_{x \rightarrow 0} \frac{x^{1000} - 703x^{506} + \pi x^{17} + 480}{-372x^{75} + 39x^{14} + \sqrt{7}x^5 - 24}$

$$\lim_{x \rightarrow 0} \frac{x^{1000} - 703x^{506} + \pi x^{17} + 480}{-372x^{75} + 39x^{14} + \sqrt{7}x^5 - 24} = -20$$

$$\boxed{-20}$$

(b) $\lim_{x \rightarrow 4} \frac{x^2 + 2x - 24}{x - 4}$

$$\lim_{x \rightarrow 4} \frac{x^2 + 2x - 24}{x - 4} = 10$$

$$\boxed{10}$$

(c) $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right)$

$$\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$$

$$\boxed{0}$$

最终答案:

$$\boxed{-20, 10, 0}$$

X 已验证: `numeric_limit`

题 AP2

Using a calculator, try to guess $\lim_{n \rightarrow \infty} n \sin\left(\frac{\pi}{n}\right)$. (Remember that n is in radians, not degrees!)

解:

$$\lim_{x \rightarrow \infty} n \sin\left(\frac{\pi}{n}\right) = \pi$$

最终答案:

$$\boxed{\pi}$$

X 已验证: numeric_limit

题 AP3

Draw a circle of radius 1. Inscribe a regular n -gon inside it.

未自动求解: 无法提取极限表达式

求解汇总

题号	求解器	求解	验证
Q1	sympy_limit	X	—
Q2	conceptual_template	X	—
Q3	conceptual_template	X	—
Q4	conceptual_template	X	—
P1	conceptual_template	X	—
P2	sympy_limit	X	—
P3	conceptual_template	X	—
AP1	sympy_limit	X	X
AP2	sympy_limit	X	X
AP3	none	×	—

注: 本文档由 homework-solver 流水线自动生成(确定性符号计算, AI 使用记录见 ai_usage_log.jsonl 与交付文档)。